Project Proposal (Week 2), A Short Guide

D7065E, what the Week 2 proposal must contain

Course staff

How to use this guide. It explains what to put in the **Week 2 proposal**. See the accompanying worked proposal example. Keep the proposal to **1–2 pages**.

What the proposal is, and is not

The proposal exists to agree on a **direction** and get a **green light to start building**. It is **not** the architecture: C4 diagrams, requirements tables, interfaces, and tests are *not* expected yet. Those are built **iteratively** during development and end up in the architecture document and final report (see the report-writing guide and the worked report example). At this stage all that is needed is the intended build, the reasons for it, and the open uncertainties. The design *will* change later, that is expected and good.

What it must answer

Six short questions, a paragraph or a few bullets each:

1. **Use case, and why.** Which use case, and why it was chosen.
2. **Scenario.** The one end-to-end story being aimed for (the eventual demo).
3. **What will be built.** A rough sketch of the major pieces: what senses, what decides, what acts, **how data flows through the system, where it is stored**, and what users will see. No formal diagrams required. Each major piece should run as its **own process**, communicate through defined interfaces, and connect to **BuildSim**.
4. **The autonomous part.** Roughly what decides what, and the approach to *start* with. Rule-based, optimisation, ML, or LLM are all fine, *sophistication earns no marks*, so start simple.
5. **Biggest unknowns & risks.** What is uncertain or might not work. Honesty here is the point.
6. **Physical simulation approach.** How environmental conditions and occupancy will be generated, what physical model, dataset, or hybrid approach will drive the sensor values. This makes the simulator a first-class part of the project, not an implementation detail; the next section gives examples.
7. **First step & rough plan.** The riskiest thing to prototype first to test the idea, plus a rough sense of the weeks ahead.

Include, at the top of the proposal, a link to the team's **GitHub repository** where the project will be developed.

Simulating the building

BuildSim provides the rooms but **not their physics**, so the proposal should say how the **physical processes** that produce sensor values and the **movement of people** that drives them are simulated.

- **Physical processes**, sketch how each sensor value will be generated: a simple physics-based model (for example a heat balance for temperature, or a mass balance for CO_2 that rises with occupancy and falls with ventilation), replay of a public dataset, or a hybrid. It need not be exact, it must have realistic trends and respond to the actuators so the control loop closes.
- **Movement of people (occupancy)**, occupancy usually drives everything else (heat, CO_2, lighting). Describe the pattern to model: a weekday schedule (arrivals, lunch, departures), meeting-room bookings, empty weekends, and some randomness so no two days are identical.

Examples of approaches. Simple physics, a heat balance for temperature, or CO_2 rising about 40 ppm per person per hour and decaying with ventilation. Data replay, feed a public dataset through the sensors (e.g. the UCI *Occupancy Detection* set, or *ASHRAE* building-energy data). Hybrid, replay real data with an injected event (e.g. a real temperature trace plus a simulated fire). For occupancy, a schedule such as arrive 07:00–09:00, lunch 11:30–13:00, leave 16:00–18:00, with meeting bookings and weekend gaps.

No equations are needed yet, just the intended approach.

How it is approved

The proposal is approvable when the **direction is clear and feasible**, not when the design is complete. Approval is a green light to start, not a grade. What follows is the architecture document, which grows alongside the code; a brief "changes since the proposal" note in the final report, explaining what was revised and why, is something the examiner looks for.